

FinSentinel

Financial Sentiment Analysis using LLMs

Dori Fu • Rhea Nayar • Fatema Jaynab

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github.com/rn2357-cloud/financial-sentiment-analysis

0.978

Best Macro F1

6

Model Tiers

2,264

FPB Sentences

Executive Summary

Problem

Manually tracking financial sentiment across hundreds of headlines is slow, costly, and inconsistent. Existing tools lack calibration for a retail investor's perspective.

Goal

Automatically classify financial headlines as positive, negative, or neutral from a retail investor's perspective, with structured risk metadata (type, severity, horizon).

Approach

5-tier pipeline: Majority baseline/TF-IDF+LogReg → FinBERT zero-shot → FinBERT+LoRA → LLM prompting (zero/few/CoT/RAG+CoT) → ReAct agent + DistilBERT distillation.

Result

FinBERT+LoRA achieves 0.978 macro F1 at 50ms latency and zero API cost — matching GPT-4o-mini few-shot, surpassing TF-IDF by +11 points on macro F1.

Motivation & Related Work

- Retail investors process hundreds of financial headlines daily.
- Financial language is domain-specific — 'profit warning' is not just general negativity.
- Directional signals from a retail investor differ from raw sentiment.
- Existing tools lack calibration for investor perspective.
- **Dataset: Financial PhraseBank (FPB AllAgree)**

2,264 sentences annotated by 16 finance professionals. Highest-quality subset where all annotators agreed.

Deng et al. (2023)

Reddit Market Sentiment Analysis

Method: LLM in-context learning + CoT on Reddit posts with weak labels. Fine-tuned model generalizes to FiQA.

 *Limitation: Only positive/neutral/negative — no multi-dimensional risk tagging.*

Vijayan (2023)

Prompt Structured Extraction

Method: ChatGPT / BARD with standard, persona, CoT, and few-shot prompts for structured extraction from travel emails.

 *Limitation: General domain (travel), not finance. No investor-perspective calibration.*

Our contribution: extends both papers into multi-dimensional financial risk tagging with quantitative benchmarks.

5-Tier Pipeline Architecture

Tier 0-1	Tier 2A	Tier 2B	Tier 3	Tier 4-5
Baselines	FinBERT ZS	FinBERT+LoRA ★	LLM Prompting	Agent+Distilled
Majority class + TF-IDF+LogReg	Domain-pretrained BERT (110M params)	PEFT r=8, 3 epochs 0.27% params	Zero/Few/CoT RAG+CoT (GPT-4o- mini)	ReAct agent + DistilBERT student
F1: 0.34 / 0.87	F1: 0.963	F1: 0.978	F1: 0.83-0.98	F1: 0.26 / 0.54

★ Best accuracy-cost-latency balance: FinBERT+LoRA — 0.978 F1 · 50ms · zero API cost · 0.27% params trained

Implementation & Experiment Details

Data & Evaluation		Key Hyperparameters	
Dataset:	FPB AllAgree (2,264 sentences)	LoRA rank r:	8
Train / Test:	80% train / 20% test, stratified	LoRA alpha:	16 · dropout 0.1
LLM eval:	200-sentence stratified sample	LoRA targets:	Q + V attention layers
Cross-domain:	FiQA-Sentiment (Twitter/Stocktwits)	Training:	3 epochs · lr 2e-4 · batch 16
Primary metric:	Macro F1 — 61.4% neutral imbalance	LLM model:	GPT-4o-mini, temperature=0
Stats tests:	McNemar + Wilson 95% CIs	Few-shot:	6 examples (2 per class, held-out)
API cost:	Distillation corpus: \$0.37 total	RAG:	BM25 (40%) + FAISS (60%), k=3
		Distillation:	500 examples · 5 CoT paths · MSE loss
		Student:	DistilBERT · 5 epochs · 66M params

Demo — FinSentinel: Financial Risk Intelligence

Live headlines

Fetches real-time Yahoo Finance RSS headlines by ticker (AAPL, MSFT, NVDA, etc.)

Sentiment pipeline

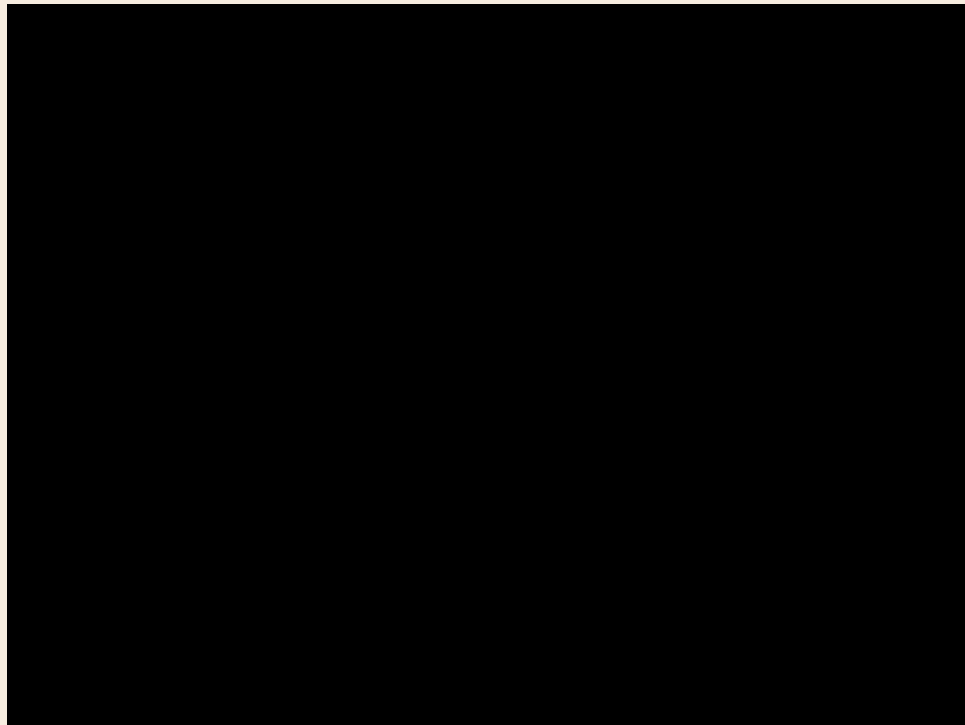
Classifies each headline with zero-shot / few-shot / CoT / RAG+CoT prompting

Daily scores

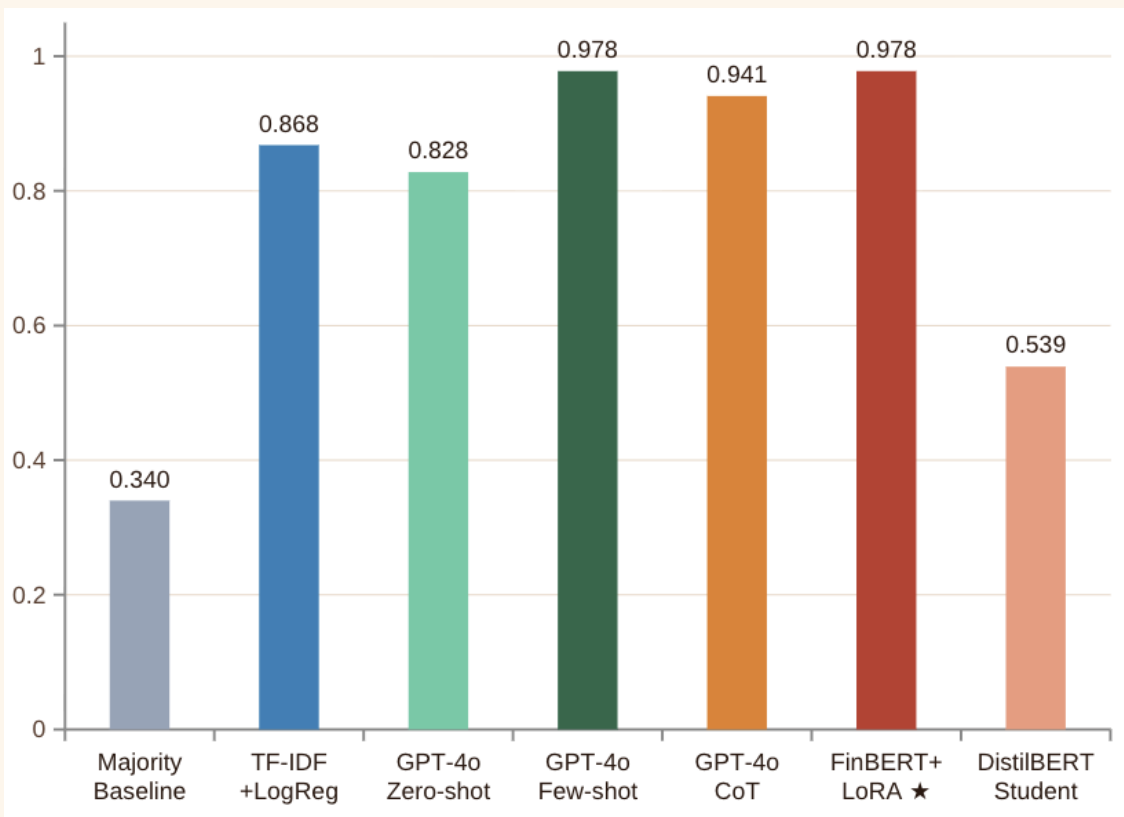
Aggregates per-headline sentiment into daily scores per ticker (range: -1 to +1)

Price correlation

Fetches actual yfinance returns and computes Pearson + Spearman correlation



Results — Model Performance



FinBERT+LoRA ties GPT-4o few-shot (0.978 F1) at zero API cost and 50ms latency

Few-shot beats zero-shot by +15 F1 points ($p < 0.001$, McNemar)

CoT (0.941) underperforms few-shot — adds conservative neutral bias

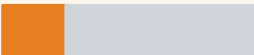
FiQA cross-domain: accuracy drops to 10% — clean benchmark $\neq$ real-world

Results — Error Analysis & Statistical Tests

Dominant Error: Neutral → Positive

Zero-shot  6/8 neutral→pos

Few-shot  3/5 neutral→pos

CoT  1/7 neutral→pos

 Neutral→Pos  Other errors

Root cause: zero-shot sees 'revenue' or 'sales' and infers positive without reading direction. Few-shot calibrates this by teaching investor perspective.

Statistical Significance

$p < 0.001$

Zero-shot → Few-shot
(McNemar, $n=200$)

$p = 0.109$

Few-shot → CoT
(not significant)

27 / 29

Disagreements fixed by
few-shot vs zero-shot

+10.6 pts

Macro F1 gain over
TF-IDF baseline

Conclusion

01 Domain beats scale

FinBERT zero-shot (0.963) outperforms GPT-4o-mini zero-shot (0.828) despite being 70× smaller. Domain pre-training matters more than model size.

02 Few-shot calibration is the biggest lever

6 labeled examples boost macro F1 from 0.828 → 0.978 ($p < 0.001$). They teach the model to judge directional investor signals, not surface words like 'sales'.

03 FinBERT+LoRA is the practical winner

Matches GPT-4o-mini few-shot (0.978 F1) at zero API cost, 50ms latency, and 0.27% of params trained. Best accuracy-cost-latency tradeoff.

04 Real-world gap remains

Strong FPB results (98.5%) collapse on FiQA (10%). Future work: domain adaptation, larger distillation dataset, live sentiment-price RSS dashboard.

Complete Model Benchmark (FPB)

Model	Accuracy	Macro F1	Latency	API Cost	Note
Majority baseline	64.5%	0.340	—	Free	Lower bound
TF-IDF + LogReg	90.3%	0.868	<1ms	Free	Keyword-based
FinBERT zero-shot	96.0%	0.963	~80ms	Free	Strong domain model
GPT-4o-mini zero-shot	—	0.828	~1.2s	Paid	Neutral-bias errors
GPT-4o-mini few-shot	97.5%	0.978	~1.2s	Paid	p<0.001 vs zero-shot
GPT-4o-mini CoT	93.0%	0.941	~1.5s	Paid	Overcorrects to neutral
RAG + CoT	—	0.953	~2.0s	Paid	Marginal RAG gain
ReAct Agent	—	0.260	~5s+	Paid	Parsing failures
FinBERT+LoRA ★	98.5%	0.978	~50ms	Free	Best overall ✓
DistilBERT student	78.4%	0.539	~10ms	Free	Neg class recall=0.02

FiQA cross-domain: all models drop sharply (accuracy ~10%). Strong FPB results do not transfer to informal microblog financial text.